

CS383

Software Design Document

(SDD)

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1. Introduction

In response to the global challenges of food waste and hunger, the Food Preservation Mobile Application was conceptualized to streamline the donation process of surplus food and support communities in need.

This document presents the key architectural and design decisions that guide the development of the application. It highlights ethical considerations, describes user activities, illustrates the overall system structure, and explains how data flows across the system using Data Flow Diagrams.

Furthermore, it defines class relationships within the system and represents system behaviors and interactions using standardized UML diagrams.

2. Ethics:

1- **Data Security and Privacy:** The application must prioritize the security and privacy of user data by implementing robust encryption techniques to protect sensitive information, such as user profiles, donation details, and messages, ensuring unauthorized access is prevented.

2- **Empowerment:** The application should empower charitable organizations to manage food distribution for vulnerable groups (such as needy families) with dignity and efficiency. Food will first be delivered to the organizations, which will then distribute it to beneficiaries in a manner that preserves their dignity.

3- **Social Responsibility:** The app should positively contribute to society by reducing food waste. It should streamline the process of collecting surplus food and delivering it, helping food donors and charities coordinate efficiently.

4- **Efficiency:** The application should be highly efficient in delivering surplus food to charitable organizations as quickly as possible, by automating tasks such as matching donors with charities based on data analysis.

3. Activity Model

Agile Model

The Agile Model is an iterative and flexible software development approach that emphasizes collaboration, continuous feedback, and adaptive planning. It is particularly suitable for Barakah App, where frequent interaction with Users (such as donors, Volunteers, and administrators) and rapid evolution of features are necessary to meet changing user needs and operational demands.

Why Choose the Agile Model?

- **Incremental Delivery:** The system is built in small, functional units known as Sprints, with each Sprint delivering a specific set of features. For example, one Sprint may focus on user authentication and role selection, while another may implement donation scheduling or location based pickup features. This allows for continuous integration of new functionality without compromising stability.
- **Continuous Feedback and Improvement:** Agile enables early and frequent delivery of working components, allow testing and evaluate the system during development. This feedback loop is essential in refining both the user experience and the backend logic. For instance, donors may suggest improvements in the donation management interface, which can then be implemented in subsequent Sprints.
- **Prioritized Feature Development:** Agile supports flexible prioritization of features. Core functions such as donation creation and request approval can be developed first, while optional features like data analytics or historical reports can be planned for later iterations, depending on user demand and project timeframes.
- **Enhanced Collaboration:** Regular meetings between developers and stakeholders ensure better alignment with real world expectations. The project team can adapt the development plan based on real usage scenarios, ensuring that the final product delivers maximum value.
- **Risk Reduction and Rapid Testing:** By breaking development into smaller cycles, issues are identified early and resolved before expanding the system. Since testing is integrated within each Sprint, quality assurance becomes an ongoing process, reducing the likelihood of critical errors in the final product.

Examples of Agile Sprints:

Sprint 1: User registration, login, and role-based access.

Sprint 2: Donation request submission and location selection.

Sprint 3: Managing incoming requests and volunteer interaction.

Sprint 4: Notification system and user communication tools.

Sprint 5: Administrative controls and data validation mechanisms.

Sprint 6: Reporting and donation tracking dashboard.

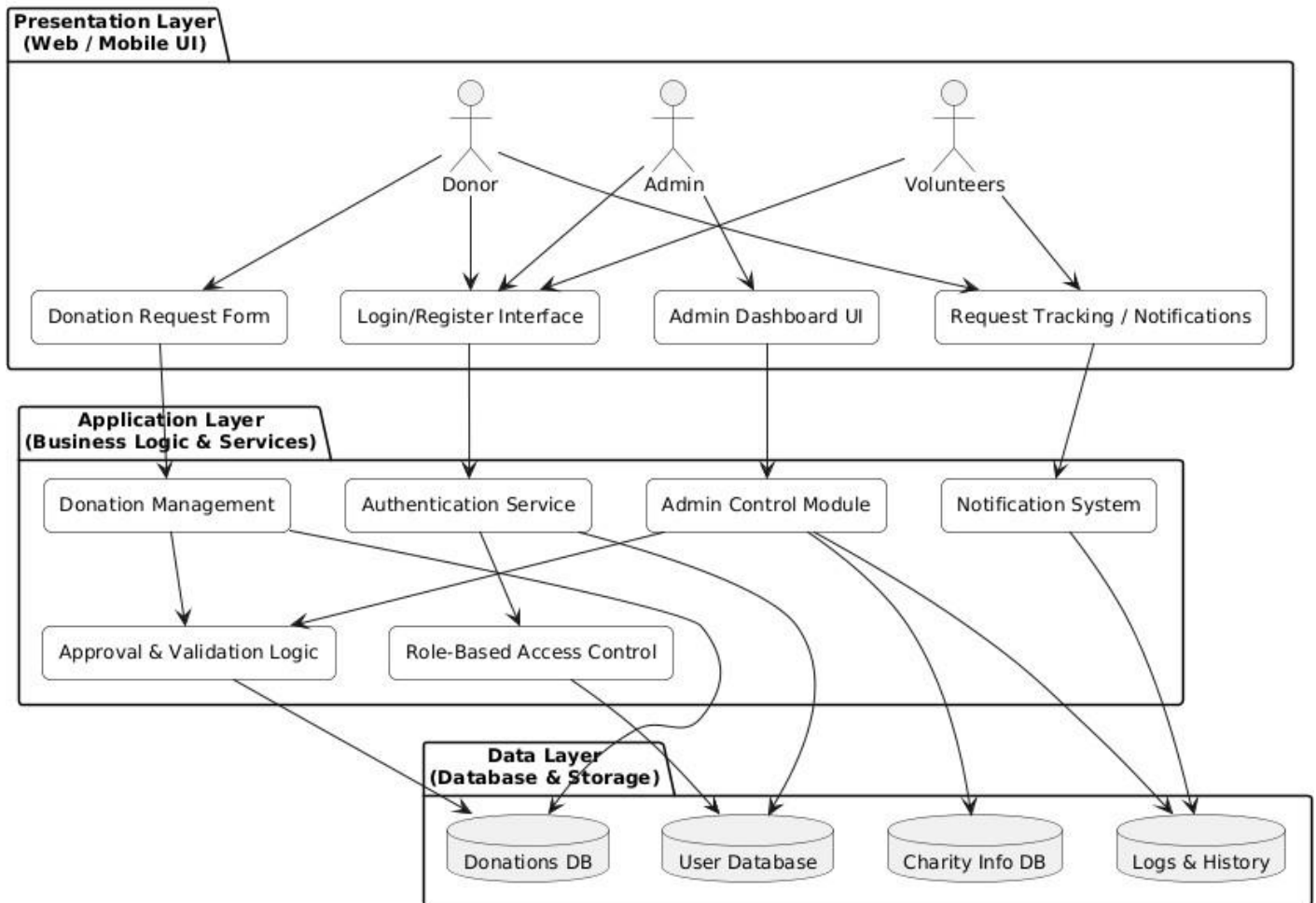
Activity Steps in Each Sprint:

1. **Requirement Analysis** – Identify user stories and define Sprint goals based on feedback and system priorities.
2. **Design** – Develop UI/UX wireframes and system architecture specific to Sprint features.
3. **Implementation** – Write code and integrate new components with the existing system.
4. **Testing** – Conduct unit, integration, and user acceptance tests for new features.
5. **Deployment** – Release the Sprint deliverable for review or real time usage.

This cycle is repeated for each Sprint, allowing the system to grow gradually into a complete and user driven platform for managing food donations effectively.

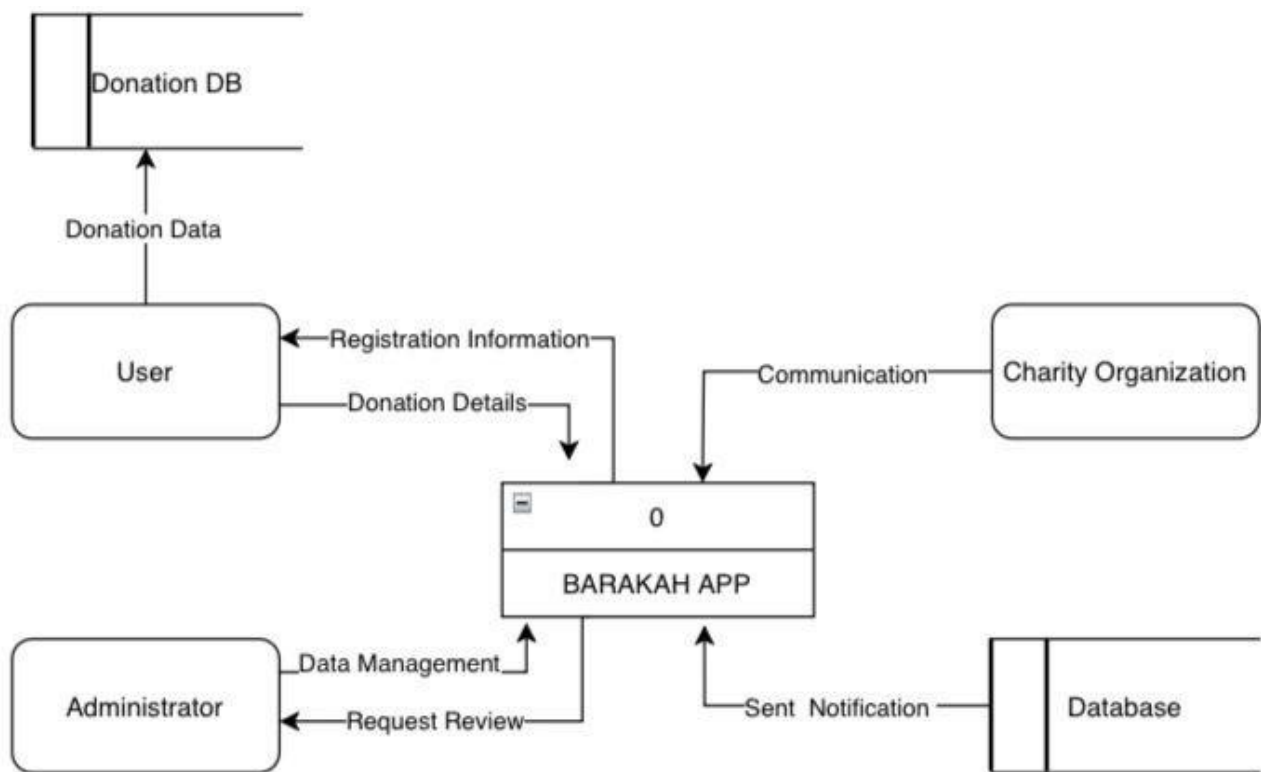
4. Architecture Model:

Barakah App - System Architecture



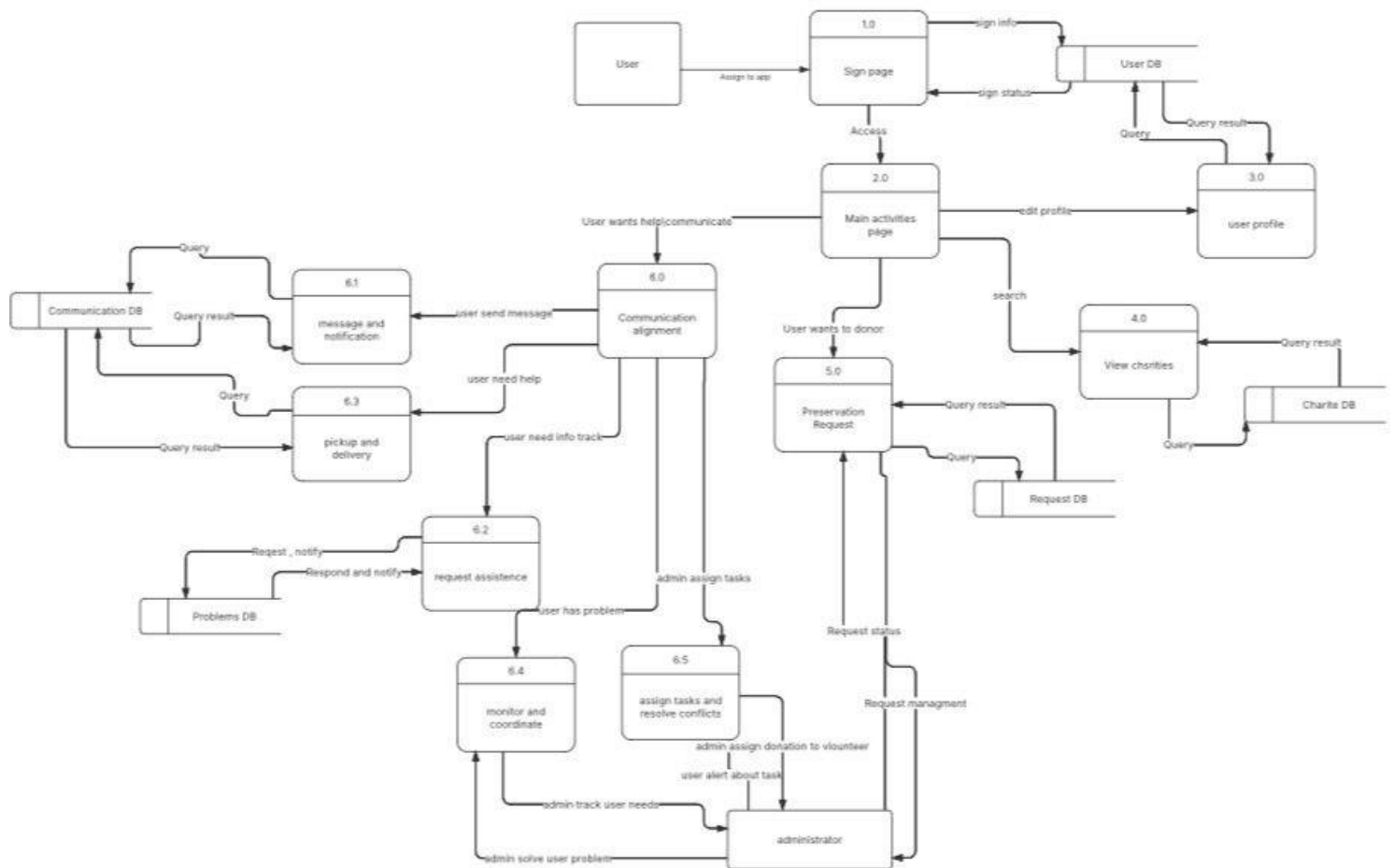
5. Data Flow Diagram

5.1 DFD Level 0

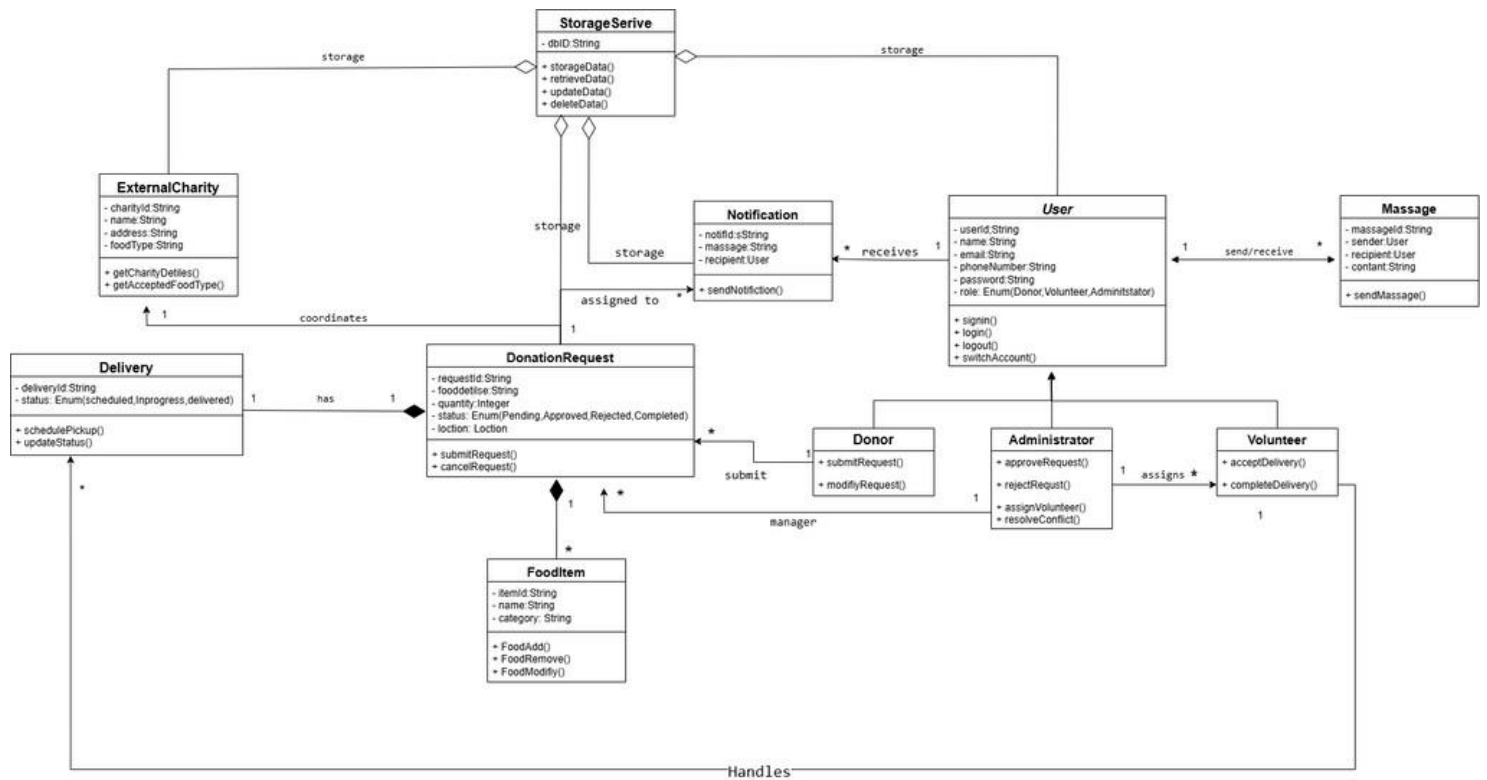


5. Data Flow Diagram

5.2 DFD Level 1

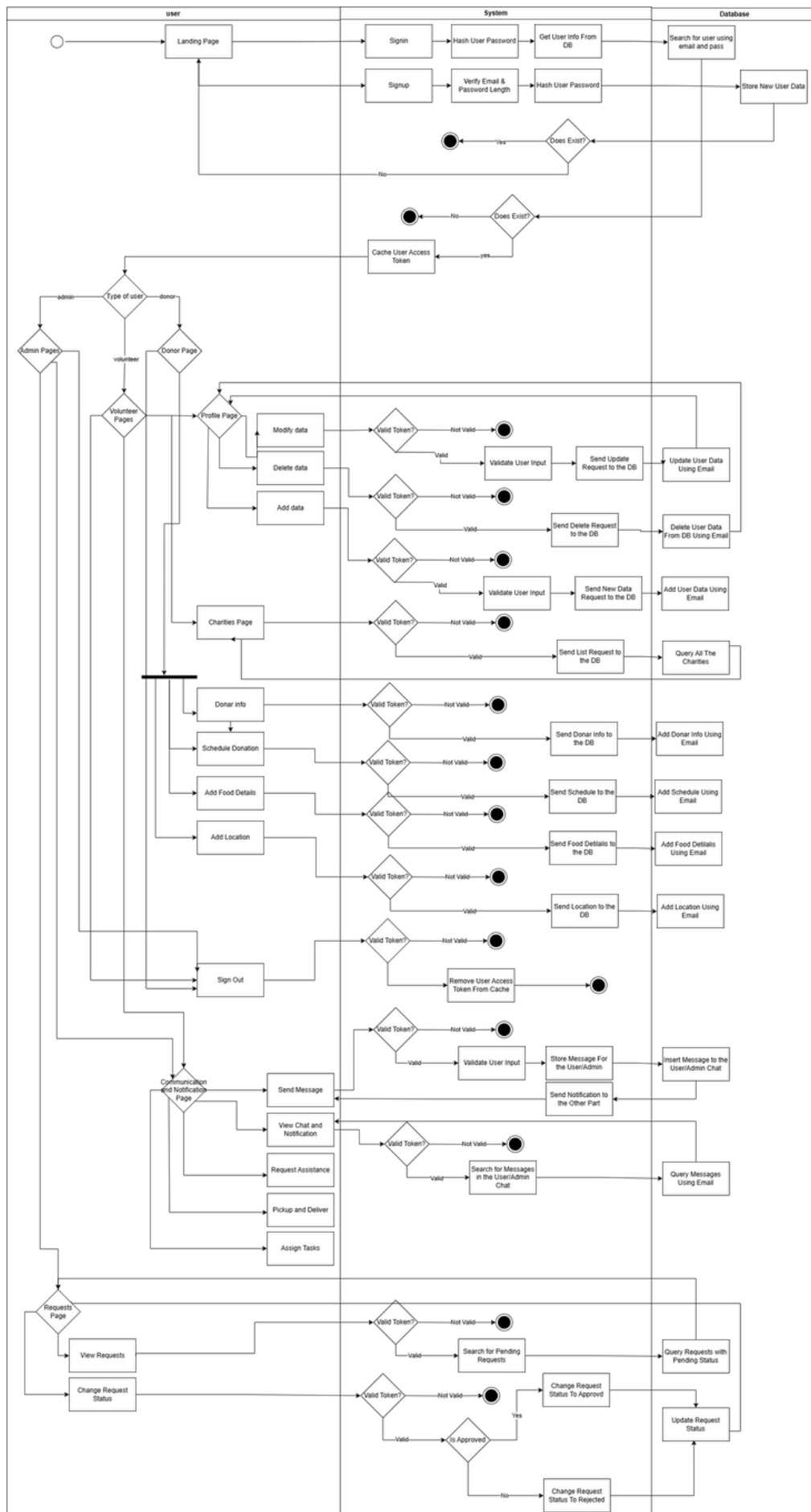


6. Class Diagram



5. UML Diagrams:

5.1 : Activity Diagram



Activity Diagram Description

The activity diagram illustrates the flow of an invoice and donation management application, beginning with the user's choice to log in or sign up. Upon launching the app, users are prompted to either log in if they already have an account or create a new one. The system verifies the submitted information during sign-up, and upon successful validation, the user is directed to the homepage.

There are three types of users in the system: Admin, Volunteer, and Donor.

For Donors:

On the Donor page, users can view, add, edit, or delete their data. Donors can also browse charitable organizations and communicate with the admin. Additionally, they can add a donation request, which is a key function. A logout option is provided for when the user wishes to exit the session.

For Volunteers:

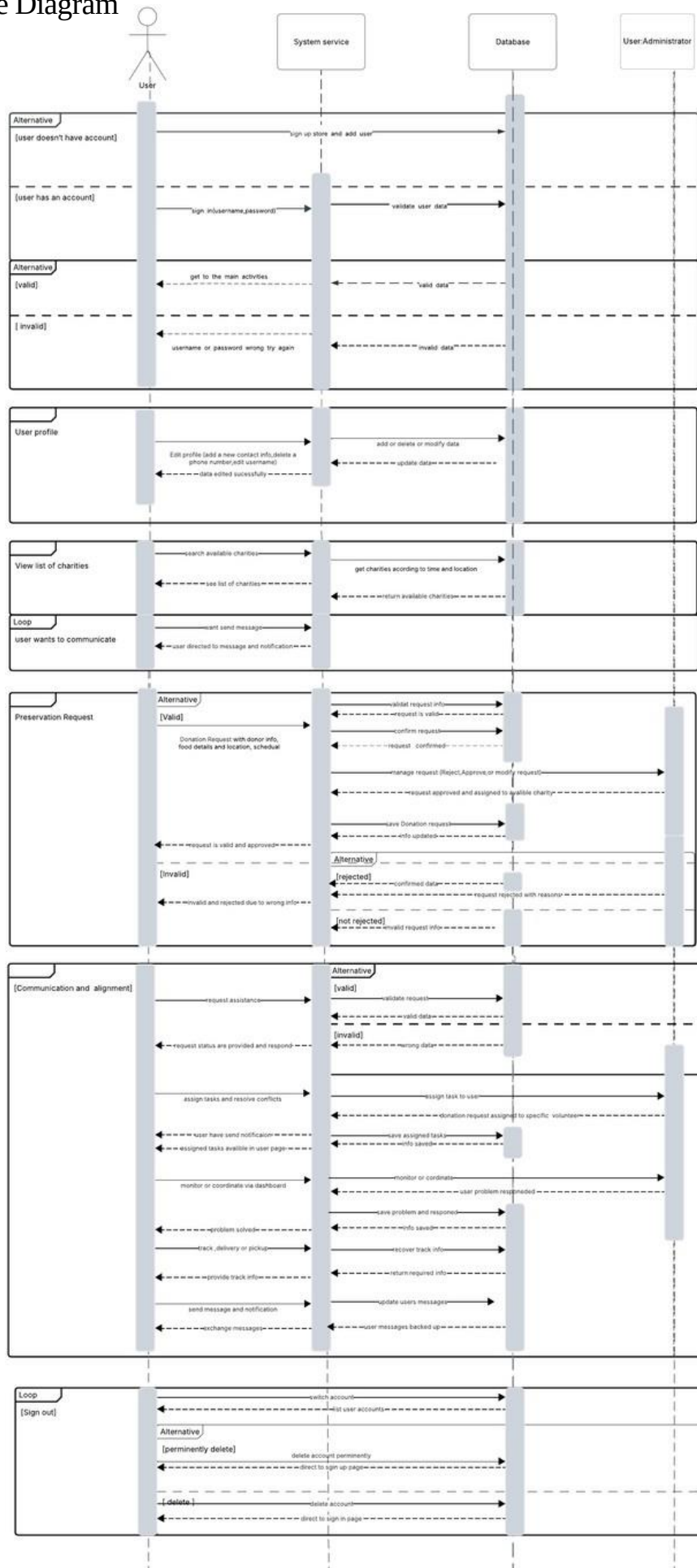
The Volunteer has access to a dedicated page where they can input their personal information, location, and food details. They can also view donation requests and select a request to deliver. They are also provided with a logout option.

For Admins:

The Admin is able to contact volunteers and access the request page, where they can accept, reject, or modify any incoming request. They can also view volunteer data and view donor data. Like other users, they have the option to log out.

5. UML Diagrams:

5.2 : Sequence Diagram



5. UML Diagrams:

5.3 : State Diagram

